

Final Project

SOF3720 : Introduction to Artificial Intelligence

Type: Group Project (2–3 students)

1. Objective

The purpose of this project is to apply AI techniques learned in the course to solve a real-world problem, design and implement an intelligent system, analyze performance and evaluate results, demonstrate software engineering best practices, and reflect on ethical and professional implications of AI systems.

2. Project Requirements

A. Problem Formulation (10%)

- Clearly define the problem.
- Identify state space / input features.
- Identify actions (if applicable).
- Identify constraints.
- Define cost / objective function.
- Justify why AI is needed.

B. AI Algorithm Implementation (15%)

Implement at least one core AI technique such as Search (BFS, DFS, UCS, A*), CSP, Minimax with Alpha-Beta, Genetic Algorithms, Reinforcement Learning, Decision Trees, KNN, Naive Bayes, Regression, Bayesian reasoning, or Monte Carlo simulation.

Pure library-only implementations are not allowed. Students must clearly explain algorithm logic.

C. System Design (15%)

- Include UML diagrams (Use Case + Class Diagram minimum).
- Provide system architecture description.
- Explain data flow.

- Ensure modular design.

G. Final Demonstration (10%)

- Live system demonstration.
- Clear explanation of algorithm.
- Ability to answer technical questions.

3. Deliverables

1. Project Proposal
2. Source Code (GitHub link required)
3. Technical Report (PDF)
4. Presentation Slides
5. Final Demo